

HDOC DAY-12

QUESTION:

Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

- Find the sum and product of the first and last digits. **E.g.** Number: 12345, Sum = 6, Product = 5
- Count how many digits are lesser than, equal, and greater than a given digit `k`. **E.g.** Number: 12345, k=3, Count_lesser=2, Count_equal=1, Count_greater=2

Algorithm:

1. Start.
2. Input a positive integer n.
3. Display menu:
 - 1: Sum and product of first and last digits
 - 2: Count digits less than, equal to, and greater than k
4. Input user's choice.
5. Use switch-case.

For choice 1:

1. Last digit = $n \% 10$.
2. Repeatedly divide a copy of n by 10 until one digit remains.
3. Remaining digit is the first digit.
4. Find sum and product.
5. Display result.

For choice 2:

1. Input digit k (0 to 9).
2. Repeatedly take last digit using $\% 10$.

3. Compare it with k.
4. Increase less, equal, or greater count.
5. Divide number by 10 and repeat until it becomes 0.
6. Display counts.

C program:

```
#include <stdio.h>
```

```
int main() {
```

```
    long long n, temp;
```

```
    int choice, first, last, k, digit;
```

```
    int sum, product;
```

```
    int less = 0, equal = 0, greater = 0;
```

```
    printf("Enter a positive integer: ");
```

```
    scanf("%lld", &n);
```

```
    if (n <= 0) {
```

```
        printf("Invalid input! Please enter a positive integer.\n");
```

```
        return 0;
```

```
    }
```

```
    printf("\nMenu\n");
```

```
    printf("1. Find sum and product of first and last digits\n");
```

```
    printf("2. Count digits less than, equal to, and greater than k\n");
```

```
printf("Enter your choice: ");
```

```
scanf("%d", &choice);
```

```
switch (choice) {
```

```
    case 1:
```

```
        last = n % 10;
```

```
        temp = n;
```

```
        while (temp >= 10) {
```

```
            temp = temp / 10;
```

```
        }
```

```
        first = temp;
```

```
        sum = first + last;
```

```
        product = first * last;
```

```
        printf("First digit = %d\n", first);
```

```
        printf("Last digit = %d\n", last);
```

```
        printf("Sum = %d\n", sum);
```

```
        printf("Product = %d\n", product);
```

```
        break;
```

```
    case 2:
```

```
        printf("Enter digit k (0 to 9): ");
```

```
        scanf("%d", &k);
```

```
if (k < 0 || k > 9) {  
    printf("Invalid digit k!\n");  
    return 0;  
}
```

```
temp = n;
```

```
while (temp > 0) {  
    digit = temp % 10;
```

```
    if (digit < k)
```

```
        less++;
```

```
    else if (digit == k)
```

```
        equal++;
```

```
    else
```

```
        greater++;
```

```
    temp = temp / 10;
```

```
}
```

```
printf("Count_less = %d\n", less);
```

```
printf("Count_equal = %d\n", equal);
```

```
printf("Count_greater = %d\n", greater);
```

```
break;
```

default:

```
    printf("Invalid choice!\n");  
}
```

return 0;

```
}
```

```
#include <stdio.h>  
  
int main() {  
    long long n, temp;  
    int choice, first, last, k, digit;  
    int sum, product;  
    int less = 0, equal = 0, greater = 0;  
  
    printf("Enter a positive integer: ");  
    scanf("%lld", &n);  
  
    if (n <= 0) {  
        printf("Invalid input! Please enter a positive integer.\n");  
        return 0;  
    }  
  
    printf("\nMenu\n");  
    printf("1. Find sum and product of first and last digits\n");  
    printf("2. Count digits less than, equal to, and greater than k\n");  
    printf("Enter your choice: ");  
    scanf("%d", &choice);  
  
    switch (choice) {  
        case 1:  
            last = n % 10;  
            temp = n;  
  
            while (temp >= 10) {  
                temp = temp / 10;  
            }  
  
            first = temp;  
            sum = first + last;  
            product = first * last;  
  
            printf("First digit = %d\n", first);  
            printf("Last digit = %d\n", last);  
            printf("Sum = %d\n", sum);  
            printf("Product = %d\n", product);  
            break;  
  
        case 2:  
            printf("Enter digit k (0 to 9): ");  
            scanf("%d", &k);  
  
            if (k < 0 || k > 9) {  
                printf("Invalid digit k!\n");  
                return 0;  
            }  
    }  
}
```

```

    if (k < 0 || k > 9) {
        printf("Invalid digit k!\n");
        return 0;
    }

    temp = n;

    while (temp > 0) {
        digit = temp % 10;

        if (digit < k)
            less++;
        else if (digit == k)
            equal++;
        else
            greater++;

        temp = temp / 10;
    }

    printf("Count_less = %d\n", less);
    printf("Count_equal = %d\n", equal);
    printf("Count_greater = %d\n", greater);
    break;

default:
    printf("Invalid choice!\n");
}

return 0;

```

Compilation and execution:

```

(kali㉿kali)-[~]
$ vi h12.c

(kali㉿kali)-[~]
$ gcc h12.c -o h12

```

Test case table:

Test case	Input number	choice	k	Expected result
1	45678	1	-	First=4, last=8, sum=12, product=32
2	70009	1	-	First=7, last=9, sum=16, product=63

3	24680	2	5	Less=3, equal=0, greater=2
4	55555	2	5	Less=0, equal=5, greater=0
5	10101	2	1	Less=2, equal=3, greater=0

```

Enter a positive integer: 45678

Menu
1. Find sum and product of first and last digits
2. Count digits less than, equal to, and greater than k
Enter your choice: 1
First digit = 4
Last digit = 8
Sum = 12
Product = 32

(kali㉿kali)-[~]
$ ./h12
Enter a positive integer: 70009

Menu
1. Find sum and product of first and last digits
2. Count digits less than, equal to, and greater than k
Enter your choice: 1
First digit = 7
Last digit = 9
Sum = 16
Product = 63

(kali㉿kali)-[~]
$ ./h12
Enter a positive integer: 24680

Menu
1. Find sum and product of first and last digits
2. Count digits less than, equal to, and greater than k
Enter your choice: 2
Enter digit k (0 to 9): 5
Count_less = 3
Count_equal = 0
Count_greater = 2

(kali㉿kali)-[~]
$ ./h12
Enter a positive integer: 55555

Menu
1. Find sum and product of first and last digits
2. Count digits less than, equal to, and greater than k
Enter your choice: 2
Enter digit k (0 to 9): 5
Count_less = 0
Count_equal = 5
Count_greater = 0

```

```
(kali㉿kali)-[~]  
$ ./h12  
Enter a positive integer: 10101  
  
Menu  
1. Find sum and product of first and last digits  
2. Count digits less than, equal to, and greater than k  
Enter your choice: 2  
Enter digit k (0 to 9): 1  
Count_less = 2  
Count_equal = 3  
Count_greater = 0
```